

2023

Bachelor in Business Administration

Course Title: **Business Statistics**

Code No: **STT201** Semester: **3rd Semester**

Full Marks: **60** Pass Marks: **24** Time: **staff of a bank of 50 employees was found to be normally distributed with mean of Rs 1200 and standard deviation of Rs 120. Find the probability of employees having income (a) between Rs 1050 to Rs 1400 (b) between Rs 1250 to Rs 1500.19.From the following distribution, find the coefficient of skewness.**

Also comment the result.Monthly income (Rs 000)Number of workers0-10015100-20050200-30075300-40040400-50030500- 6001020.From the following marks distribution of students of a campus, calculate the coefficient of kurtosis. Also, interpret the result.MarksBelow 1010 - 2020 - 3030 - 4040 - 5050 & aboveNo. of students50100150906050Section "D"Comprehensive Answer Questions:[20]21.Following table shows the income and expenditure of people of certain locality of small town city of Nepal.Income (000Rs)Expenditure (000 Rs)6022552054155217482552215316471749185019Find out:a. Coefficient of variations of income and expenditure and interpret the results.b. Correlation coefficientc. Two regression lines.d. Estimate the expenditure of a person whose income is Rs 70 thousand.

Candidates are required to answer the questions in their own words as far as practicable.

Section A

Brief Answer Questions:[10 × 1 = 10]

1. The covariance of X and Y is 30 and the variances of X and Y are 25 and 64 respectively. Find Karl Pearson's coefficient of correlation between X and Y.
2. If the values of upper quartile and lower quartile are 63 and 59 respectively, then calculate the coefficient of quartile deviation.
3. The equations of two regression lines are $4X-5Y+33=0$ and $20X-9Y-107=0$, find the mean values of X and Y.
4. Given that $P(A \cap B) = 0.7$, $P(A) = 0.4$ and $P(B) = 0.5$, then find out the value of $P(A \cup B)$.
5. Find $P(X=4)$, the mean of Poisson distribution (λ) is 1.5.
6. Calculate the standard error of mean when population size (N) = 1000, sample size (n) = 100 and standard deviation (σ) = 10.
7. In a moderately asymmetric distribution, the values of mean and mode are 20 and 23 respectively. Compute the value of median.
8. Calculate coefficient of variation, when values of mean and standard deviation are 32 and 17 respectively.
9. Given that value of correlation coefficient is 0.84, interpret the result on the basis of coefficient of determination.
10. List out any four types of random sampling techniques.

Section B

Short Answer Questions: (Attempt any FIVE Questions)[5 × 3 = 15]

11. The following table shows the marks distribution of students in a subject:

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	10	25	45	20	18	12

Find the lowest marks of the top 20% of the students.
12. A random sample of 50 items gave mean weight of 7.5 kg and a standard deviation of 1.5 kg. Find 95% confidence limits of weight within which the population mean would lie.
13. The following table represents the marks of 100 students:

Marks	0-20	20-40	40-60	60-80	80-100
Students	14	27	15	Find the missing frequencies when modal marks is 48.	
14. From the following income distribution of 500 workers in a locality, find the median value of the income distribution.

Income (000 Rs)	No. of workers
Below 20	50
20-40	90
40-60	150
60 - 80	100
80 - 100	60
100 & above	50
15. From the following income distribution, calculate the appropriate measures of dispersion.

Daily income (Rs)	Number of workers
Below 50	10
50-100	18
100-150	25
150-200	20
200-250	16
250 and above	11
16. The following information was obtained from two investment companies A and B:

Company	A	B
Average return (000Rs)	28	37
Variance	25	36
No. of observations	100	100

Calculate combined standard deviation.

Section C

Long Answer Questions: (Attempt any THREE Questions)[3 × 5 = 15]

17. The following table shows the average life of electric bulbs manufactured by X and Y companies:

Company	X	Y
No. bulbs	n ₁ 100	100
Mean life in hours	1300	1250
Standard deviation	80	90

Test whether there is any significant difference in mean life of the electric bulbs produced by two companies at 5% level significance.
18. Daily income of the part-time staff of a bank of 50 employees was found to be normally distributed with mean of Rs 1200 and standard deviation of Rs 120. Find the probability of employees having income (a) between Rs 1050 to Rs 1400 (b) between Rs 1250 to Rs 1500.
19. From the following distribution, find the coefficient of skewness. Also comment the result.

Monthly income (Rs 000)	Number of workers
0-100	15
100-200	50
200-300	75
300-400	40
400-500	30
500- 600	10
20. From the following marks distribution of students of a campus, calculate the coefficient of kurtosis. Also, interpret the result.

Marks	Below 10	10 - 20	20 - 30	30 - 40	40 - 50	50 & above
No. of students	50	100	150	90	60	50

Section D

Comprehensive Answer Questions:[20]

21. Following table shows the income and expenditure of people of certain locality of small town city of Nepal.

Income (000Rs)	Expenditure (000 Rs)
60	22
55	20
54	15
52	17
48	25
52	21
53	16
47	17
49	18
50	19

Find out:
a. Coefficient of variations of income and expenditure and interpret the results.
b. Correlation coefficient
c. Two regression lines.
d. Estimate the expenditure of a person whose income is Rs 70 thousand.